3.Perform basic Image Handling and processing operations on the image is to read an image in python and Convert an Image to show outline using Canny function.1. Aim

1. Aim

To write and execute a Python program to read an image and detect its outlines using the Canny Edge Detection function in OpenCV.

2. Procedure

1. Import the OpenCV library using import cv2.
2. Read the input image using cv2.imread().
3. Check whether the image is loaded successfully.
4. Convert the image into grayscale using cv2.cvtColor().
5. Apply the Canny function using cv2.Canny() to detect the edges.
6. Display the original image and the outline image using cv2.imshow().
7. Press any key to close the image windows.

3. Code

```
import cv2


# Read the image
img = cv2.imread("image.jpg")


# Check whether image is loaded
if img is None:
    print("Error: Image not found!")
else:
    # Convert image to grayscale
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)


    # Detect edges using Canny
    edges = cv2.Canny(gray, 100, 200)


    # Display original and outline images
    cv2.imshow("Original Image", img)
```

```
cv2.imshow("Image Outline", edges)

# Save the outline image
cv2.imwrite("outline_image.jpg", edges)

cv2.waitKey(0)
cv2.destroyAllWindows()4.Sample input:
```



5.sample output:



4. Result

Thus, the Python program to read an image and detect its outlines using the Canny function in OpenCV was executed successfully.